Clinical Risk Assessment Report: Patient 38

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 12.8% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.58x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which reduced the patient's absolute risk by 1.9%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Patient Age (Age) **Value: 53.0 [Bottom 16%] (+1.9%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.62 [Avg (75th %ile)] (+1.6%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

3. Protective / Mitigating Factors (Reducing Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.66 [Bottom 17%] (-1.9%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.1 [Avg (67th %ile)] (-1.4%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -0.67 [Avg (27th %ile)] (-1.2%)**

Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.43 [Bottom <1%] (-1.2%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.06 [Avg (42th %ile)] (-1.0%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.07 [Avg (68th %ile)] (-0.9%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Machine Learning Feature Attribution (SHAP Decision Path)